The Sudden Infant Death Syndrome

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THE SUDDEN and unexpected deaths of apparently thriving infants represent an appalling tragedy to families of victims and are a major cause of infant mortality. Such deaths apparently have been occurring throughout recorded human history, yet only in recent years has this tragic condition been recognized as a specific clinicopathologic entity. Since the majority of such deaths are accompanied by only trivial anatomic changes, there has been a plethora of hypotheses put forth to fill the factual void as to the specific cause of death. Because of widespread, recent attention to the problem, a rather bulky literature on the subject has been developed, much of it weighted toward theory rather than observed fact. Many individuals with "theories in search of diseases" have applied these to sudden infant death, with results that have varied from stimulating productive research on the one hand to the production of tragic guilt reactions on the part of familes of victims on the other.

The goal of this paper is to present a general review of the problem, including data from 500 consecutive, personally studied cases since January 1, 1965. During this period the author has been a member of a research team involved in a multifaceted study of the problem and has been responsible for the conduct of all postmortem studies. These cases were drawn from two parallel sources: the King County Study, representing an intensive case-finding effort and including every possible case occurring in this given geographic area,[1-3] and the previously unreported Washington Study in which random cases of sudden infant

NOTE: This work was supported in part by U.S. Public Health Service Grant H.D. 01659; an appropriation (1963–1966) from the Washington State Legislature; and the Washington Association for Sudden Infant Death Study, the Sudden Infant Death Memorial Fund and the Laura Patience Paschall Memorial Fund.

death were brought to us for postmortem examination from adjacent counties in Washington State. At the time of this writing 526 infants have been entered in the combined study, and data for this paper are based on the first 500.

The subject of sudden infant death represents an enormous challenge to the scientist, and an equal challenge to those concerned primarily with the delivery of humanitarian care to families of victims. These deaths leave in their wake an intense and potentially catastrophic series of reactions, among which guilt and blame are uppermost. The physician caring for the dead infant not uncommonly participates in this guilt reaction ("What did I miss?"), and this may interfere with his ability to counsel effectively. The counselor, armed with inaccurate information or erroneous bias as to the cause of sudden infant death, is in a position to enhance the guilt-blame reaction of parents greatly, while the fully informed individual can intervene with dramatic effectiveness in preventing these cruel and unnecessary reactions. It is hoped that the present review will prove useful both to the scientist capable of advancing our understanding of the cause of these tragic deaths, and to the physicians and other health care professionals concerned primarily with care of the family.

DEFINITION AND TERMINOLOGY

One of the more difficult and controversial problems in this entire subject is that of definition. As is often the case with diseases or syndromes, a rigid definition is neither possible nor even appropriate until the etiology and pathogenesis are understood. Any sizeable series of sudden, unexpected infant deaths will include a number of cases in which generally acceptable causes for death will be uncovered by postmortem study. Among 500 infants in the Washington study, 75 (15%) were thus categorized (Table 1).

The remaining 425 cases (85%) not listed in Table 1 were classified as "unexplained." This includes many cases listed among the "possibly explained" group in the 1969 Conference.[4] Many of the difficulties in classification of "grey zone" cases were discussed in that paper and will not be repeated here. It should be emphasized, however, that equivocal, atypical or "grey zone" cases comprise only a small proportion of those left after conventionally acceptable causes of death, such as those in Table 1, are excluded. Since respiratory inflammation was the largest category of explained death, the criteria were the same as those employed for the left-hand column of Table 26 in the 1969 Conference[4]; namely, gross, suppurative consolidation of lung tissue and/or purulent, material-filling lumen of lobar bronchi, mainstem bronchi or trachea, with distal atelectasis. Lesser degrees of pneumonia or tracheobronchitis, requiring the microscope for detec-

TABLE 1.—CAUSES OF DEATH IN 75 CASES EXPLAINED BY AUTOPSY

	No. of Cases
INFECTION (52 cases)	
Pneumonia, bronchiolitis	24
Meningococcemia	8
Gastroenteritis, dehydration	5
Viremia, myocarditis	5
Meningitis	2
Epiglottitis	1
Beta-strep sepsis	1
H. influenzae sepsis	1
Miscellaneous	5
TRAUMA, ACCIDENTS (11 cases)	
Head injuries	9
Accidental suffocation	2
MALFORMATIONS (8 cases)	
Cardiovascular	4
Hydranencephaly	1
Incarcerated inguinal hernia	1
Internal hernia, volvulus	1
Urethral valves, hydronephrosis	1
MISCELLANEOUS (4 cases)	
Status asthmaticus	2
Coronary periarteritis	1
Endocardial fibroelastosis	1

tion or confirmation, were deemed inadequate to serve as a conventional, a priori cause of death.

A tremendous number and variety of *lesions* were found among the remaining 425 cases, including many minor inflammatory changes, six small, clinically occult neuroblastomas, hemodynamically insignificant cardiac malformations and several instances of underlying, genetically determined disease of great significance to the families, but not adequate to serve as an explanation for sudden death. The value of the postmortem examination in sudden, unexpected infant death is beyond question, despite the fact that some 85% of cases are left "unexplained" by the examination.

Our current working definition of the sudden infant death syndrome (SIDS), then, is that derived from the 1969 Conference:

> "The sudden death of any infant or young child, which is unexpected by history, and in which a thorough postmortem examination fails to demonstrate an adequate cause for death."[5]

This definition has several weaknesses, many of which were discussed at the Conference. It undoubtedly embraces a number of disease entities yet to be discovered. It has, however, the virtue of not ordaining a specific set of positive findings which must be present to justify a diagnosis of SIDS. The major problem with this definition in our opinion is that it is by necessity an *exclusional* definition, and thus conveys a rather misleading impression about SIDS. Anyone who has examined even a small series of such cases will be impressed by the similarities in clinical setting and in the postmortem findings in the vast majority of SIDS cases thus defined, and most workers soon come to the conclusion that *SIDS is, with rather uncommon exceptions, a diagnosis that can be made with considerable certainty on the basis of positive findings.* These will be emphasized in the later sections of this paper.

It is necessary, before proceeding, to insert a few words on the problem of terminology. In the 1969 Conference,[5] the author suggested the application of a uniform term for this clinicopathologic entity, SIDS. That suggestion has been quite generally accepted. Some workers have chosen alternative terms, some of which are unimportant modifications of SIDS. We feel most emphatically that the name used for this condition should not increase the problems of those charged with the counseling of families. For example, "unexplained" as part of the name, when appearing on death certificates or in the newspaper obituary column, carries with it the inappropriate implication that "they really don't know why the baby died." This weakens the case of a counselor trying to convey to the parents the concept that their infant died of a recognized, albeit imperfectly understood, condition.

Another term that often leads to difficulty is "pneumonitis," or any other label implying the baby died of an infection or other conventional disease process. Such terms too often are misunderstood as implying that the baby may have had symptoms which, if detected and treated in time, would have prevented death.

Less troublesome, but also suboptimal in our opinion, is "sudden unexpected death in infancy" or other similar terms, such as sudden unexpected death or sudden unexpected infant death. These names have the disadvantage of including deaths such as those in Table 1, which also were sudden and unexpected. The term SIDS has had decided advantages, in our experience, in counseling families, as it conveys the impression of a definite condition. "Crib death" (or "cot death") is an appropriate and useful euphemism and one we use frequently in discussing this condition.

HISTORIC NOTE

A definitive history of SIDS remains to be written and is beyond the scope of this essay. However, it is important to determine whether this is in fact a disease of modern civilization, which would have a profound bearing on theories of pathogenesis. The evolution of attitudes toward this condition is also of exceptional interest and serves as a condemnation of both ancient and contemporary approaches to a frequent human tragedy.

For millennia, unexpected deaths of infants during sleep were routinely attributed to overlaying by mother or wet-nurse. An often quoted instance of this interpretation is to be found in I Kings 3:19–20:

> "And this woman's child died in the night; because she overlaid it. And she arose at midnight, and took my son from beside me, while thine handmaiden slept, and laid it in her bosom, and laid her dead child in my bosom."

Sudhoff, cited by Garrison,[6] mentions a German placard, dated 1291, recognizing the danger of suffocating infants by overlaying and forbidding mothers from taking to bed with them infants under 3 years of age.

That deaths of this type did not achieve prominent mention in major pediatric works of early times perhaps is not surprising. Stupendous infant and childhood mortality rates from infectious diseases, malnutrition and other currently preventible conditions completely overshadowed these events, and presumably also conditioned both parents and physician to a more casual acceptance of early death, albeit unexpected. Additionally, for most of the centuries of recorded human history, deliberate infanticide by exposure, suffocation or other means was accepted practice, as abundantly illustrated in Garrison's superb history of Pediatrics.[6] Thus, an infant death due to presumed overlaying was likely to be viewed with toleration in most societies. Garrison[6 (p. 72)] quotes from Congreve's 1695 drama, "Love for Love," in which a spendthrift libertine says of the mother of his illegitimate offspring, "She knows my condition well enough, and might have overlaid the child a fortnight ago, if she had any forecast in her."

On the basis of rather scanty and anecdotal mention of this type of death, we can only presume that SIDS was a fairly frequent phenomenon prior to the 19th century. By that time medical periodicals became numerous, and mention of sudden infant death during sleep were cited frequently.

Probably the major impetus toward development of a sizeable scientific literature on SIDS in the 19th century was the rise of the theory that the deaths hitherto routinely ascribed to suffocation were in fact attributable to an *enlarged thymus*. An excellent review of the early development of this concept is that of Lee, published in 1842.[7] One result of the thymic theory was the appearance of several articles in which a strikingly modern and humanitarian approach to the problem is seen. Deaths previously thought to be due to neglect, accident or even deliberate infanticide could now confidently be ascribed to a diagnosable medical condition if an autopsy were done, and the grieving, guilt-ridden and even legally prosecuted parent or nurse could be exonerated when an enlarged thymus was demonstrated!

Berthold's 1898 paper[8] illustrates this attitude:

> Gentlemen! It is known to you all, that not uncommonly one hears or reads of a previously healthy child found dead in his bed, and that these situations easily give rise to dreadful suspicions of gross neglect or even murder, especially when an illegitimate foster child is involved. The severest blame often descends upon babysitters, foster mothers, wet nurses, and nursemaids when this horrible catastrophe occurs, as it is usually assumed that the young life was ended by suffocation in the bedding, which could have been avoided by due prudence.
>
> Grawitz describes such a case, in which a servant girl was entrusted with the care of an infant who was sleeping in its cradle near her. In the morning the previously healthy baby was found dead in the cradle. The girl was imprisoned, and the authorities ordered an autopsy, which was performed by Liman & Grawitz. Liman was unable to disprove Grawitz' contention that death was attributable to a colossaly enlarged thymus pressing on trachea, bronchi, and vessels. The magistrate released the woman. . . .

Because of this humanistic approach, it is with mixed feelings that we note the gradual development, beginning in the early 1850's, of doubts as to the thymic nature of sudden infant deaths and their clinical counterpart, "thymic asthma." Recognition that thymuses previously viewed as enlarged were in fact no larger than in healthy infants dying accidentally, led to critical analysis of the issue, culminating in Friedleben's 1858 monograph[9] in which the issues of thymic death and thymic asthma authoritatively were denounced. Widely accepted concepts are slow to die, however, and many authors continued to report "thymic death" of infants during the succeeding decades. Until 1889 the presumption of adherents of this school was that death was due to mechanical compression of the airway, vessels or nerves by the enlarged gland. In that year, however, Paltauf,[10] in a rather impressive essay, proposed that enlargement of the thymus was but one component of a constitutional disorder in which generalized lymphoid hyperplasia, generalized arterial hypoplasia, small adrenals and postadolescent hypogenitalism were associated features. This constitutional diathesis, termed *status thymico-lymphaticus,* was observed in a number of sudden unexpected deaths in childhood and even in young adults, and was applied widely as an "explanation" of death in the next few decades. "Spasmus glottidis" was the usually assumed immediate mechanism of death.

However, with acquisition of definitive normal data on the size of the thymus in health and disease,[11] thymic explanations of death subsided into the background. With the decline of the thymic theory, blame again descended upon those caring for the dead infant. A striking example of this is Templeman's 1893 paper.[12] He stated that from 1882–1891 in the town of Dundee, Scotland, 399 infants were reported to the police as having been found dead while in bed with their parents. With an 1891 census of 153,587, this "overlaying" death rate of about 40 per year is indeed impressive and perhaps implies that SIDS was *more* frequent than at the present time. From Templeman's description of the pathologic findings, higher wintertime incidence and age distribution, it is clear that the population was composed predominantly of SIDS cases. The point of his essay was that a disproportionately large number of deaths occurred on Saturday nights, which he attributed to the frequency of alcoholic intoxication on Saturday evenings. Templeman was most bitter in his accusations of such parents and advocated the harshest possible legal prosecution in cases where parents went to bed intoxicated. Since he was police surgeon of Dundee, we only can wonder how often this attitude might have colored his courtroom testimony and led to cruel miscarriages of justice.

Though a scattering of short reports in the early decades of the 20th century voiced the opinion that sudden unexpected infant deaths during sleep might be due to natural causes, these efforts had negligible influence on the general attitude that suffocation was the mechanism of death.

In the opinion of the present author, the "modern era" of SIDS research was ushered in by the landmark papers of Werne and Garrow.[13–16] In a series of carefully documented papers appearing between 1942 and 1953, these authors presented compelling evidence for operation of natural, probably inflammatory mechanisms in many of these deaths, and discounted suffocation. It should be noted that Werne and Garrow's work was not done at a major teaching or research center, but at the Office of the Chief Medical Examiner, Borough of Queens, New York. For some time thereafter, the major contributions to the literature emanated from the offices of coroners or medical examiners. Among these, the efforts of Dr. Adelson and colleagues in Cleveland are worthy of special mention.[17] Without the dedication of these and other coroners and medical examiners, working under adverse conditions in understaffed, underfunded circumstances, SIDS research could never have arrived at its presently promising state, and attitudes would have remained reminiscent of the Dark Ages.

Work in the last few years has accelerated considerably, and at the present time SIDS generally is recognized as a definite disease entity by a majority of forensic and pediatric workers, despite a lack of agreement as to its precise definition, terminology and causation.

There has been a gratifying acceleration in the number of scientific publications on the subject and in the awakening of a general awareness of the subject on the part of academic and public service agencies. It is not unrealistic to hope that in the near future significant new answers to the cause of SIDS will be forthcoming, and effective help for families now too often subjected to neglect and even prosecution generally will become available.

EPIDEMIOLOGY

INCIDENCE

Nearly all studies reported to date have revealed incidence figures in the range of 2–3 per 1,000 live births.[18] These figures pertain solely to contemporary, predominantly urbanized, Western civilization. If a mean figure of 2.5 per 1,000 live births is applied to the current annual birth rate of about 3.5 million, it is apparent that nearly 10,000 SIDS deaths occur yearly in the United States.

Sudden infant death syndrome is the largest single cause of post-neonatal infant mortality, accounting for approximately one third of all deaths in infants between 1 week and 1 year of age. During the third and fourth months of life this condition apparently accounts for more than half of all infant deaths (Table 2).

Peterson's study[19] of SIDS incidence in the climatically dissimilar cities of Seattle, Washington, and San Diego, California, revealed identical rates.

Probably the best data comparing urban with rural incidence are those of Froggatt *et al.*[20, 21] They observed a higher incidence in Belfast than in the rural counties of Ulster; but when the metropolitan areas within these rural counties were pooled and compared with the remaining areas, no differences were observed. Presently available data suggest that differences between urban and rural areas are not great,

TABLE 2.—RELATIVE SIDS MORTALITY
(King County, Washington, 1968–1971)

MONTH OF LIFE	TOTAL DEATHS*	SIDS DEATHS Number	SIDS DEATHS Percent
2	125	58	46
3	106	57	54
4	81	43	53
5	53	18	33
6	31	11	35
7–12	95	16	17
Total	491	203	41

*Courtesy of D. R. Peterson, M.D.

and those that do exist may reflect socioeconomic differences in the populations being compared, rather than urban-rural differences per se. It would be instructive to compare SIDS incidence in upper and middle class rural regions with, for example, rural Appalachia. It also would seem important to determine the incidence of SIDS in cultures and climates that differ widely from the Europeanized temperate regions where existing data have been obtained.

Age Distribution

One of the most consistent and perhaps the single most characteristic feature of reported series of SIDS cases is the age distribution. The sparing of very young infants, peak incidence between 2 and 4 months, and rapid decline before the age of 6 months are common to virtually all studies. We are aware of no other condition with this unique age distribution. Table 3 summarizes the age at death of 425 infants in the Washington SIDS study.

It is apparent from Table 3 that over half of SIDS events have occurred by 3 months of age and 91% occur in the first 6 months of life. Although no infant in our series was over 1 year of age, we have reviewed several cases sent for consultation from other centers that appeared to represent during the first half of the second year of life. Only five (1%) occurred during the first 2 weeks of life, despite vigorous attempts at case finding for the King County cases that comprise over three fourths of the total series. These five cases occurred at 4, 9, 10 and 13 (two cases) days, respectively. This relative sparing of the very young infant is real, therefore, and must be reconciled with any proposed explanation for SIDS.

We are unable to demonstrate that gestational length is related sig-

TABLE 3.—Age Distribution
(425 SIDS Cases)

Age (Months)	Number	Percent
0–1	19	4
1–2	107	25
2–3	113	27
3–4	73	17
4–5	49	11
5–6	31	7
6–7	11	
7–8	8	
8–9	7	
9–10	3	9
10–11	3	
11–12	1	
Over 12	0	

nificantly to age at death, so premature infants seem to die at approximately the same age as term infants.[3] However, data relative to this point are currently being reworked by Dr. D. R. Peterson and a definitive answer on this fundamental point should be forthcoming in the near future.

Circumstances of Death; Relationship to Sleep

Some of the literature on SIDS is misleading with respect to circumstances of death and other epidemiological features, because *sudden unexpected death,* regardless of cause, was the entity being investigated.[17] Cases of the type included in our Table 1 were grouped together, therefore, with those that currently would be defined as SIDS. So, it has not been recognized generally that SIDS is a phenomenon that almost invariably occurs unobserved during periods when the infant almost certainly is asleep.[1, 3, 20] *No* SIDS event was observed in our series. All deaths occurred during normal sleeping periods, with the exception of one in which the victim was seen playing happily in an infant "stroller" about 5 minutes before being discovered apparently lifeless. In the King County study,[3] over one third of deaths occurred with another person asleep in the same room, and several dozen times an awake adult was in an immediately adjacent area. Deaths occurred twice in car beds placed in the rear seat of an automobile with the parents in the front seat. *In none of these cases was there an audible outcry at the approximate time of death.* We conclude from these observations that SIDS probably is related importantly to sleep, and that death is not associated with an outcry.

On the other hand, despite its apparent silence, there is evidence that death may be attended by vigorous motor activity. Bedclothes may be clutched in the hands of the dead infant, or may be pulled up over his head. He may be found in a peculiar position, wedged in the corner of the crib, may have the face pressed through the slats or may have fallen between the mattress and wall. Very frequently the face will be straight down in the bedding, with pressure pallor involving the center of the face, a position certainly not representing a normal sleeping position. Any of the above circumstances almost routinely lead to suspicion that the infant died by suffocation. The arguments against this theory will be given later, but it is worthy of mention at this point that the epidemiologic and pathologic findings are identical in infants found with the face entirely free and those with the face covered.

We have been unable to demonstrate that the position in which the infant was placed (back, side or abdomen) differs from normal sleeping habits in this community.[22]

Many theories of SIDS tend to incriminate factors related to the sleeping environment. Hypothermia, hyperthermia, recent climatic changes and suffocation in bedding are among these. There currently is no evidence that the seasonal peaks in SIDS incidence (to be discussed below) are related specifically to cold weather. For example, the wintertime incidence of SIDS in Seattle is identical to that in San Diego.[19] Furthermore, we have seen no increase in incidence during spells of unusually cold weather. Steele[23] found an apparent correlation with recent change in temperature and wind velocity, but in a subsequent study[24] this association was insignificant. Richards and McIntosh[24a] found no influence of mean or minimum air temperature on SIDS incidence.

SEASONAL VARIATION

Nearly all workers have observed seasonal variations in SIDS incidence, with fewer cases occurring in the summer months.[25] This fluctuation is independent of minor monthly variations in birth rate,[3, 20] and the monthly incidence curve varies markedly from year to year. Because of the latter phenomenon, seasonal peaks become somewhat blurred as the number of years surveyed increases. This seasonal influence apparently is independent of thermal influences also, since Peterson[19] found identical seasonal trends in San Diego and Seattle for the years 1962–1964, using the same ascertainment method.

We have observed an apparent tendency toward temporal clustering of cases,[3] but have been unable to prove this statistically. Froggatt's scholarly analysis[20, 21] for clustering of cases also yielded negative results, but both series have suffered from small numbers relative to the analytical methods used. Steele[23] also observed an excessive number of days on which two or more SIDS events occurred.

To summarize, SIDS seems to be affected by seasonally variable factors other than ambient temperature. There is no consistent month or season of peak occurrence in consecutive years. Clustering of cases in time and space may well occur, but has not yet been proved.

SOCIOECONOMIC FACTORS

The several studies directed toward this difficult parameter have yielded quite consistent results, with an increased incidence of SIDS in lower income groups.[3, 20, 26, 27] Houstek's intriguing data[28] suggest that SIDS incidence may decrease in a given district with elevation of standard of living. Despite its greater attack rate among the underprivileged, SIDS is in fact no respecter of class or social position. Our series includes infants born to four physicians, two dentists and a number of wealthy and prominent families in the region.

It is not clear to what extent the influence of socioeconomic factors is independent of other variables, such as race and low birth weight. Froggatt's pioneering efforts,[20,21] as multivariant analysis, represent the best available information on this point, and his results are inconclusive due to small numbers.

SEX

Virtually every series has shown a male preponderance. Among the 425 cases reported in this paper, 272 (64%) were male infants. During the same time period 52% of births in King County were male. The presumption is that this reflects the general male preponderance in mortality and infectious disease morbidity in infants, although specific factors predisposing the male to SIDS might exist.

BIRTH WEIGHT

An increased risk of SIDS among low birth-weight babies is shown by virtually all studies. In Peterson's[29] and our[3] series, for example, approximately 20% of the infants had a birth weight below 2,500 Gm, without obvious evidence of being small for dates. Peterson showed a consistently declining risk of SIDS with an increasing birth weight. Froggatt's[20,21] and our[3] data suggest that the age at death is approximately the same for the prematurely born infant as for the one born at term. This, if substantiated by further analysis, carries with it the implication that predisposition to SIDS is more intimately associated with extrauterine environmental factors than with absolute, postconceptional age.

We are indebted to Froggatt[20,21] for attempting to separate mathematically the effect of birth weight from that of social class and other variables. Within given social classes, mean birth weights tended to be lower than for matched controls of the same class, but the large numbers of cases required for multifactorial analyses have frustrated efforts at conclusive identification of the magnitude of this effect. Two generalizations, however, would seem in order: (1) any proposed theory must be consistent with an enhanced risk for the prematurely born infant, and (2) the disproportionate numbers of prematures in any SIDS series must be taken into account in analyzing certain other epidemiological factors, such as incidence of breast-fed versus bottle-fed infants, as discussed below.

MULTIPLE BIRTHS

Among our 425 SIDS cases, eight infants were members of twin pairs, of which two were members of a twinship of like sex but unknown zygosity, both of whom died of SIDS one week apart. An additional case was one of triplets.

A number of apparently simultaneous SIDS events have involved both members of twin pairs. Geertinger found five such pairs in his files,[30] of which four pairs were of like sex and one unlike. Simultaneous deaths of dizygous twin pairs also were reported by Cooke and Welch[31] and by Froggatt.[20, 21] Simultaneous twin deaths in which the sex was unstated were reported by Kraus and Borhani,[32] Wöckel and Raue[33] and Harbitz.[34] Kraus and Borhani also mention a nonsimultaneous death of both members of a twinship. On the other hand, eight papers contain reference to a total of 185 sets of twins in which only one died of SIDS.* We may add our six twinships to this, in which only one member died.

Carpenter[35] calculated an SIDS attack rate of 2.2 per 1,000 twin births, compared to 1.4 per 1,000 singleton births. Kraus and Borhani[32] presented risk figures of 8.33 per 1,000 for triplet births, 3.87 per 1,000 for twin births and 1.46 per 1,000 random births. These risk figures are not sufficient to prove an effect other than lower birth weight in multiple births. However, the occurrence of apparently simultaneous deaths in twins, regardless of their zygosity, is an intriguing phenomenon and one that suggests environmental influences may be of extreme importance in SIDS.

Parental Age and Parity

No clear impact of these factors is apparent, though several authors note an increased risk in mothers under 20 years of age[20, 25, 29, 32] and a reduced proportion of firstborn babies,[20, 24, 28, 29, 32] although the risk does not clearly increase with increasing parity beyond two. Froggatt's multifactorial analysis[20] discriminated a significantly increased risk when there was a combination of young maternal age with increased parity. It remains to be seen whether this phenomenon is independent or linked to the socioeconomic influence on SIDS. Paternal age is linked in the same way as maternal: we encountered more fathers than expected under the age of 25 years.[3]

Feeding Histories

Great confusion has been perpetrated in this area. Several early studies indicated that SIDS never occurred in breast-fed babies and several theories have been advocated on the basis of this concept. However, the King County study[3] revealed that 34% of SIDS victims had been at least partially breast-fed, and seven infants were said never to have received cow's milk in any form. Houstek's report[28] included

*References 17, 20, 28, 30, 31, 35, 36, 38.

32 totally breast-fed infants, and Froggatt's detailed studies revealed virtually identical feeding patterns in cases and controls.[20, 21] Therefore, it is concluded that, despite its other advantages, breast feeding neither prevents nor reduces the risk of SIDS.

GENETIC FACTORS

We have maintained, for purposes of practical genetic counseling, that SIDS is not a genetic disease, and that the risk of recurrence is negligible. We do emphasize, however, that SIDS is a fairly frequent disease, and that random risk figures will result in an occasional intrafamilial recurrence, just as one family may suffer losses in separate automobile accidents. The information in the next paragraphs has not altered that approach materially.

The data from twins, discussed above, is of little value in discriminating genetic from environmental factors, especially since data on zygosity are sparse. It is clear, however, that several of the simultaneous events in twins have involved dizygous twins.

Sibship and other intrafamilial recurrence of SIDS have been reported by several authors (Table 4).

Froggatt[20] presents the only data we are aware of in which the SIDS risk to siblings of cases is presented. His recurrence rate in siblings was between 11.1 and 22.1 per 1,000 siblings at risk, a rate of 4–7 times the random risk. Even if the higher figure is correct for purposes of genetic counseling, an approximately 2% recurrence risk cannot be considered a significant deterrent to having more children. Currently, a study of the families in our series is being undertaken to determine

TABLE 4.—REPORTED CASES OF SIBSHIP RECURRENCE (Excluding Twins)

AUTHOR, REFERENCE	NUMBER OF CASES	COMMENTS
Froggatt[20, 21]	4–6 families with 2 cases 1 family with 3 cases	2 families had multiple apneic spells
Steinschneider[37]	1 family with 5 possible cases	Multiple apneic spells; 1 sib 28 mo.; 2 not autopsied
Siwe[36]	4 families with 2 cases	
Müller[36]	1 family with 3 cases	
Adelson & Kinney[17]	2 families with 2 cases	
Coe & Hartman[38]	1 family with 2 cases	
Houstek[28]	1 family with 2 cases	
Vaughan[39]	1 family with 2 cases	
Valdes-Dapena[25]	1 family with 2 cases	
Porter[40]	1 family with 2 cases	
Beckwith (present report)	6 families with 2 cases	

denominator figures for number of subsequent siblings born to SIDS families, and hopefully more accurate risk figures may emerge.

Despite the apparently enhanced risk figures for siblings of SIDS cases, this is clearly not a Mendelian trait. We are aware of no report of excessive consanguinity rates among parents. The figures given above are far below Mendelian recurrence rates. Weinberg and Purdy[41] observed a variety of chromosomal abnormalities in 10 of 17 cases of SIDS, but these were of uncertain significance.

Among our 425 SIDS cases, there were three sibships in which two died and both were autopsied by the author. Of these, one set was the twin pair cited earlier. Another set consisted of Caucasian males who died at the ages of 6 and 12 weeks, respectively, with findings typical of SIDS, although a number of dissimilar, minor and physiologically insignificant dysmorphic features were encountered in both. The third sibship pair were males of Caucasian-Chinese parentage, dying at 1 month and 4½ months, respectively, with findings typical of SIDS.

Four additional cases in our study had siblings that probably died of SIDS, but were autopsied elsewhere. One of these had been hospitalized multiple times for apneic spells prior to dying of apparently typical SIDS. A subsequently delivered sibling later pursued an identical course. This is most intriguing in view of Froggatt's[20] observation that two of his familial recurrences also had multiple apneic-cyanotic spells. Steinschneider's family with recurring apneic spells and five infant deaths adds weight to Froggatt's implication that such spells might be associated significantly with enhanced sibship recurrence.

Data on other possibly affected family members are scanty. The King County study includes, among 170 cases with adequate data, 11 histories suggestive of SIDS in the sibling of one parent, or in a first cousin of the case

Several hypotheses might be proposed to account for the observations given above.

1. *SIDS might be influenced by polygenic factors.* This concept of predisposition by virtue of an "unbalanced" gene set reasonably is consonant with the above observations. The history of recurring apneic spells might be one manifestation of a polygenically determined predisposition, reflecting "autonomic instability," or an altered threshold to lethal and sublethal episodes.

2. *Some familial "SIDS" cases in fact may represent other genetically determined diseases.* As was stated in the discussion of "Definition and Terminology," it is probable that cases currently defined as SIDS will in the future be found to include hitherto undiscovered disease entities. Perhaps, for example, the familial apnea cases referred to above are clues to a specific disease entity differing in some ways from SIDS in general.

3. *Environmental factors in the family setting may be responsible for some recurrences.* Valdes-Dapena[25] studied one family with three adopted infants of genetically different backgrounds, who died of SIDS. We have studied one similar family in which two adopted sibs of different origins both died in separate events. The data on simultaneous deaths of dizygous twins cited earlier also may be interpreted in favor of the operation of as yet unknown environmental triggers.

To summarize, it would appear that a moderate tendency toward familial aggregation of cases exists, but not to such a degree as to significantly discourage the family of a SIDS victim from having a subsequent child. One possible exception to this might be the family in which there is a history of SIDS and *also* infants with severe, recurring apneic spells. The recurrence risk in the latter setting may be unacceptably high, although exact figures currently are unavailable.

Infectious Disease

It has been suspected by most workers, since the pioneering observations of Werne and Garrow,[13-16] that infectious agents played a significant role in the pathogenesis of SIDS. The age distribution, peaking at a time of maximal susceptibility, as well as the seasonal variations, coupled with pathological evidence of inflammatory infiltrates, albeit minor, have all pointed in this direction. Histories of minor illness in the days preceding death have been noted by many authors.[3, 20, 42]

In general, efforts to demonstrate *bacterial* pathogens have been unsuccessful. However, some authors, such as Sutton and Emery,[42] have found an increased isolation rate of pathogens in families of SIDS victims. Their evidence led these authors to propose a possible synergistic effect of viral and bacterial infection. Bendig, *et al.*[43] found abnormal numbers and distribution of enteric bacteria in 21 of 29 cases. Kerenyi and Fekete[44] found pathogens in 70% of 40 consecutive cases. Bacteriologic studies in general have been hampered by the fact that bodies of SIDS victims typically lie in a warm room for some time before discovery, and an appreciable interval between death and necropsy can alter relative bacterial populations in the body drastically.

Early unsuccessful attempts to demonstrate *viral* agents have been supplanted by several successful efforts.[1, 45] Significantly higher isolation rates for a variety of respiratory and enteric viruses usually associated with relatively minor diseases have been documented. We[45] found potentially pathogenic viral agents in 37.5% of SIDS cases as compared to 16.2% of controls. It seems clear that viral agents play some role in the pathogenesis of SIDS, and apparently no single agent is responsible.

We have sought for evidence of antenatal infection without success.[46]

In 15 cord bloods collected from infants subsequently dying of SIDS, with matched controls, there was no evidence of elevated IgM (mean level, 12.5 mg/100 ml compared to 15.2 mg/100 ml for controls). Ray[47] found no increase in viral isolation rates among 100 family members representing 31 SIDS families, as compared to 80 persons from 21 control families.

Of great importance is the fact that successful isolation of viral agents in our study has been from the rhinopharynx and stool, and *not* from blood or the multiple organs sampled. Thus, there is no direct evidence for massive, overwhelming viremia in these cases.

PATHOLOGIC FINDINGS

While relatively unrewarding in terms of demonstrating a "cause" for SIDS, pathologic findings do afford considerable insight into the possible mechanisms of death. In addition, the impressive repetitiveness of a narrow spectrum of minor changes lends great weight to a unitarian concept of SIDS.

External examination reveals the typical SIDS victim to be in a normal state of nutrition and hydration. Somatotypes seem to represent a full cross section of infancy, with the great majority of infants being normally plump. Over half of cases exhibit frothy fluids in the mouth and nostrils, indicative of pulmonary edema. This fluid is commonly tinged with blood, and when it has stained the bedclothes, parents often fear their infant suffered a massive internal hemorrhage. The diapers usually are wet and full of stool. The latter finding is paralleled by an empty, contracted urinary bladder and rectum in the great majority of cases. This consistent emptying of bladder and bowel contents at or near the time of death indirectly points to a rather cataclysmic agonal event, and stands in contrast to the variable bladder and bowel contents seen in routine hospital deaths. In many cases, the hands will be found to be clutching fibers resembling blanket material. Emesis will sometimes be present on the face.

Outstanding among the internal findings are *intrathoracic petechiae* (Fig. 1). Typically, these dot the surfaces of lungs, pericardium and thymus, and frequently also involve the parietal serosal surfaces of the chest. They are present in about 87% of infants in our series[4] and their regional limitation to the thorax is impressive. The latter point is well shown in the case of the thymus which is partly within and partly without the thorax. The cervical lobes are often spared, especially on the dorsal surface where the left innominate vein may provide a "damping" of intrathoracic negative pressure (Figs. 2 and 3). We have discussed previously[4] the various postulated mechanisms accounting for these petechiae, and the fact that their distribution strongly suggests that Tar-

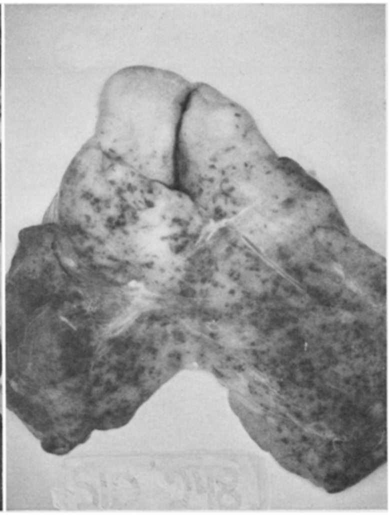

FIG. 1 (*left*).—Intrathoracic petechial hemorrhages. Note that the parietal surface of the pericardium and diaphragmatic pleura also have a few petechiae seen near the bottom.

FIG. 2 (*right*).—Dorsal view of thymus, showing sparing of cervical lobes.

dieu's original theory may have been correct. They may indicate that intrathoracic negative pressure was elevated sufficiently to cause capillary rupture during the agonal period. Most workers will agree that these petechiae are characteristic of most SIDS cases where they are seen more consistently than in any other condition, but most ascribe them to nonspecific agonal anoxia. The present writer has never been able to visualize how anoxia could be limited so regionally in its effects! It is probable, however, that in the case of the lungs, at least, much of the extravasation of blood occurs after death, since pulmonary petechiae are often distributed in a dependent fashion. Perhaps in the low-pressure pulmonary circuit there often is insufficient time during the agonal period for postrhexis hemorrhages to develop before circulation ceases. It should be emphasized, however, that diffuse splinter hemorrhages in the lungs are not at all uncommon, and are in fact the presumed source of the blood that tinges the edema fluid often seen externally.

Other typical findings include *pulmonary congestion and edema*, along with minor microscopic *inflammatory infiltrates* in the lung and upper airway. The pulmonary infiltrates are characteristically interstitial lymphocytic infiltrates of minor density. Alveolar macrophages commonly are seen. The lung changes are inadequate entirely to serve as a conventionally acceptable cause of death, but in the absence of other lesions many workers still view SIDS as being due to "interstitial pneumonitis."

We have discussed elsewhere[48] the occurrence of peculiar areas of inflammation and fibrinoid necrosis in the vocal cords. These extraordinary lesions have been quite consistent in serially sectioned larynges from SIDS cases, but also were seen with some frequency in controls. We leave open the possibility that these foci bear some relation to the pathogenesis of SIDS, but certainly they are not diagnostic of this condition.

Inflammatory changes of minor degree may be found in other organs, but usually are lacking. The adrenals show no stress effects, and in our experience are normal in weight for age. The thymus usually is plump and minimally affected by stress, although an infant occasionally dies of SIDS after recently having recovered from another illness, and in this setting the gland may be small. Thymic atrophy, however, usually is a signpost to a significantly stressful underlying disease, and most infants with atrophied glands are found to have died of other conditions than SIDS. Geertinger[30] emphasized the frequency of anomalies of parathyroid glands, consisting of absence, displacement or fusion with thymus. Two subsequent papers by others[49, 50] effectively have refuted that observation by demonstrating that the incidence of these lesions is equally high in controls.

We have seen no repetitive pattern with respect to fullness of the stomach—a wide range from empty to distended is observed. The presence of some curd in the trachea often is misinterpreted as evidence of death by aspiration. Gardner,[51] in an important analysis of death by

FIG. 3.—Dorsal view of thymus. The left, innominate vein was left on this specimen. This vessel is adherent to the posterior thymic capsule and may provide a "damping" of negative pressure, interfering with its transmission into the region of the dorsal cervical lobes.

aspiration, showed conclusively that stomach contents can easily pass into the airway and even reach the alveolar level *after* death. If aspiration does in fact cause death, as it does in some neurologically handicapped individuals, one sees stomach content impacted in the airway. This picture in SIDS cases is sometimes confused by attempts at resuscitation. On the basis of personal experience, the author is convinced that death by aspiration virtually never occurs in neurologically intact individuals, whether awake or in natural sleep.

Blood is characteristically in the fluid state, and the right heart is dilated mildly.

We have been unable to detect the glomerular "lesions" described by Stowens *et al.*,[52] although the kidneys usually are rather bloodless due to ventral distribution of livor mortis in infants dying in the prone position.

In whole-lung sections, and serially sectioned larynges from over 100 cases, we have found no evidence of mast cell degranulation.

THEORIES OF CAUSATION

Before presenting our own view as to possible mechanisms, it is worthwhile to put the existing body of theory and fact into perspective. Table 5 attempts a more or less comprehensive listing of theories proposed to date, although new ideas are being published almost weekly.

TABLE 5.—THEORIES OF CAUSATION OF SIDS

EXTRINSIC THEORIES	COMMENTS
Infanticide (Mt. Sinai J. Med. 35:214, 1968)	Certainly the most malignant theory ever proposed. Inconsistent with narrow pathologic spectrum, viral isolation rates, etc.
Suffocation	See text—"Theories of Causation"
Spinal epidural hemorrhage (Am. J. Clin. Path. 49:562, 1968)	Refuted (Am. J. Clin. Path. 52:289, 1969). We have not seen these in our series.
Aspiration of gastric contents (Lancet 1:343, 1948)	See text—"Near Misses"
Climatic changes[23]	Refuted by author's own subsequent work[29] and by Peterson.[19]
Hypothermia (Lancet 1:343, 1971)	Refuted.[19, 24, 54] We have noted no increase during cold weather.

(continued)

Exogenous hyperthermia (Frequent death certificate diagnosis)	Usually supported by scant evidence. Obviously not the cause of most SIDS.
NUTRITIONAL AND METABOLIC THEORIES	
Rickets (Beitr. gericht. Med. 26:278, 1969) (Ztschr. Kinderh. 84:565, 1960)	Might be an occasional contributing factor, but obviously not a major one. Two cases of our 425 showed rachitic changes.
Selenium, vitamin E deficiency (New Zealand M. J. 71:32, 1970) (Brit. M. J. 4:559, 1971)	Refuted (J. Pediat. 81:415, 1972). Also refuted by the fact that breast-fed babies frequently die of SIDS. See text —"Feeding Histories"
Scurvy (M. J. Australia 2:292, 1970)	We have not seen osteoporosis in ribs or vertebrae. Petechiae limited to chest. Most of our cases got supplemental vitamins, including vitamin C.
Renal dysfunction[52]	Postmortem autolysis not excluded. Others, including ourselves, have not seen the lesions described. No definite amino acid or protein abnormality was found.[20, 21]
Magnesium deficiency (Lancet 2:258, 1972)	Normal magnesium levels in vitreous humor of 12 consecutive SIDS cases were found (Lancet 2:871, 1972).
Hypocalcemia (Wien. klin. Wchnschr. 74:21, 1962)	
Deficient vitamin D_3[24]	
Acidosis (J.A.M.A. 206:1446, 1968)	
Hypokalemia (Brit. M. J. 3:646, 1972)	
Subclinical aminoacidopathy, protein overload (Lancet 1:914, 1966)	
ENDOCRINE THEORIES	
Adrenal insufficiency (J. Pediat. 67:248, 1965)	Refuted by our cortisol data (Clin. Res. 18:503, 1970).
Parathyroid deficiency[30]	See text —"Pathologic Findings."
ANATOMIC THEORIES	
Conduction system abnormality (Am. J. Cardiol. 22:479, 1968)	Disputed by Valdes-Dapena's data (Am. J. Path. 70:272, 1973).
Cartilage in conduction system (Lancet 1:64, 1971)	Rare lesion.

(continued)

Aspiration of malformed uvula (Schweiz. med. Wchnschr. 97:62, 1967)	Impressive photo, but applies only to rare cases.
Aspiration of edematous uvula (München. med. Wchnschr. 52:435, 1905)	
Collapse of trachea (Arch. Dis. Childhood 45:147, 1970)	
Postinfectious dislocation of atlas Brit. M. J. 4:625, 1971)	
Enlarged thymus	See text—"Historic Note"
Choanal atresia	Refuted (J. Pediat. 81:1145, 1972).
Positional airway narrowing	
IMMUNOLOGIC THEORIES	
Anaphylaxis to intrauterine infection (Lancet 1:912, 1966)	No abnormality of IgE observed,[46] no reduction in complement and normal cord blood IgM. Our studies have not shown evidence of mast cell degranulation.
Anaphylaxis to cow's milk (J. Immunol. 3:307, 1960) (Internat. Arch. Allergy 24:214, 1964)	
Immune complex disease (Lancet 1:210, 1972)	
Reaction to immunization (Nederl. tijdschr. geneesk 108:2061, 1964) (Deutsche Ztschr. ges. gericht. Med. 56:66, 1965)	
Hyperimmunization, "anti-antibody" (J. Clin. Path. 24:736, 1971)	Failure to find evidence of anti-antibody.[46]
Anaphylaxis to nucleoprotein (Am. J. Dis. Child. 14:465, 1917)	
Hypogammaglobulinemia (J.A.M.A. 156:246, 1954)	Refuted (J. Pediat. 63:290, 1963 and Pediatrics 42:61, 1968).
Anaphylaxis to house dust mites (M. J. Australia 2:1240, 1972)	
INFECTIOUS THEORIES	
Streptococcal sepsis (New England J. Med. 211:154, 1934) (Brit. M. J. 2:1234, 1959)	
Bacterial sepsis[42, 44]	
Combined bacterial and viral infection[42]	
Coliform enteritis-endotoxemia[43] (Pediat. News 1:3, 1967)	
Staphylococcal enteritis (Deutsche Gesundh. 21:1441, 1966)	

(continued)

Bacterial pneumonia (Acta paediat. scandinav. 59:685, 1970)	
Otitis media (Deutsche med. Wchnschr. 86:2402, 1961)	
Laryngotracheobronchitis (Am. J. Clin. Path. 50:695, 1968	
Bronchiolitis (Deutsche Ztschr. ges. gerichtl. Med. 45:7, 1956)	
Desquamative pneumonia (Frankfurt. Ztschr. Path. 69:314, 1958) (*Proceedings of the 8th International Congress on Pediatrics*, 1956)	
Interstitial pneumonitis	One of the commonest theories. Major problem is minor nature of the lesions.
Virus infections	See text—"Epidemiology"
Cytomegalovirus infection (Frankfurt. Ztschr. Path. 70:409, 1960)	
NEUROLOGIC THEORIES	
Diving reflex (Tr. Am. Clin. & Climatol. A. 76:192, 1964) (Am. Heart J. 71:840, 1966) (Am. J. Dis. Child. 123:480, 1972)	
Hypersensitive carotid sinus reflex with neck extension (Canad. M. A. J. 100:968, 1969)	
Palato-glossal reflex (M. J. Australia 2:292, 1970)	
Arrhythmia due to autonomic immaturity (Am. J. Cardiol. 22:469, 1968)	
Dive reflex or laryngospasm due to aspiration of saliva (Am. J. Dis. Child. 124:788, 1972)	
Apneic spells[37]	
Laryngospasm (see text)	
Vasovagal reflex (Arch. Path. 61:341, 1956)	
Cervical cord lesion—"Ondine's curse" (J.A.M.A. 223:330, 1973)	
MISCELLANEOUS THEORIES	
Cerebral "red substance" (J. Forens. Sci. 9:157, 1964)	

(continued)

Nasal obstruction (Am. J. Dis. Child. 116:115, 1968) (Am. J. Dis. Child. 119:416, 1970) (Arch. Dis. Childhood 46:211, 1971)	One article (Am. J. Dis. Child. 120:167, 1970) cautions that phenylephrine rebound might exaggerate nasal obstruction. We (J. Pediat. 81:1145, 1972) found no x-ray evidence of nasal obstruction in 78 SIDS victims.
Fibrinoid laryngeal necrosis[48] (J. Forens. Sci. 3:503, 1958)	See text—"Pathologic Findings"
Hyperpyrexia (Lancet 1:757, 1971)	
Pulmonary thrombosis (Arch. Dis. Childhood 28:187, 1953)	
Cerebral edema (Paediat. Grenzgeb. 10:213, 1971)	
Deranged eosinophile function (Maryland M. J. 18:70, 1969)	
Chromosomal abnormalities[41]	
Maternal methadone addiction (J.A.M.A. 220:1733, 1972)	We have only two heroin-addicted mothers and no known methadone addicts in our series.
Toxic follicle reaction (Ztschr. Kinderh. 96:335, 1966)	
Vascular injury (Am. J. Path. 26:730, 1950)	
Myeloid leukemia (Brit. M. J. 4:423, 1972)	Purely speculative—one of the easiest theories to discard.
Myocardial electrolyte disturbance (Beitr. gerichtl. Med. 26:32, 1969)	
Genetic cardiac conduction defect (Lancet 2:56, 1966)	
Pulmonary surfactant deficiency (Scarpelli, E.: *The Surfactant System of the Lung* [Philadelphia: Lea & Febiger, 1968])	Cites sparse or absent surfactant layer in 20 SIDS lungs. However, the pulmonary edema fluid is clearly frothy in SIDS. Also, edema per se can decrease surfactant.

Many of the theories need no discussion. For others, comments or pertinent references supporting or refuting the various hypotheses are cited in Table 5. In this section a few theories which are most likely to be pertinent to the person counseling SIDS parents are presented in more detail.

SUFFOCATION

This is among the most persistent of theories, and is the one usually uppermost in the minds of parents. The effective counselor, therefore, must be aware of specific reasons why the facts about SIDS are incompatible with suffocation.

1. The age distribution curve of SIDS is inconsistent with this hypothesis. Why is the very young infant spared, at a time when he seems most vulnerable, yet by the time of peak incidence, between 2 and 3 months, good head control has been achieved?
2. Cases found with faces covered are clinically and pathologically identical with those found with faces entirely free.
3. The high rate of viral isolation from victims and the consistent finding of minor inflammatory disease are not accounted for by accidental suffocation.
4. Ordinary bedding is probably incapable of producing suffocation.[53]
5. The seasonal curve which might otherwise be interpreted as being due to the presence of more bedcovers, is in fact independent of specific thermal environment.[19, 24, 54]

ANAPHYLAXIS

This theory seemed extremely attractive at the time it was proposed. Subsequently it has been dealt several severe blows.

1. Cow's milk feeding histories are not constant in SIDS, and several large series have found no correlation of SIDS with feeding pattern (see section on "Epidemiology").
2. Clausen, *et al.*,[46] in the King County study, were unable to demonstrate elevations of IgE, nor was there evidence of complement consumption.
3. Carpenter and Shaddick[55] found personal and family histories of allergy identical in 110 SIDS cases and 196 matched controls.
4. Several studies[56–59] have failed to demonstrate an excess incidence of antibodies to cow's milk proteins in SIDS.
5. Valdes-Dapena and Felipe[60] found no increase in cells producing antibodies to cow's milk proteins in SIDS tissues.
6. We have found no evidence of significant degranulation of mast cells in SIDS tissues (See "Pathology" section).
7. Why does SIDS become rare after 6 months of age, while other hypersensitivity diseases continue into childhood and beyond?
8. This theory in most of its forms fails to explain the epidemiologic and microbiologic evidence for infection.

Neurological Theories

Table 5 lists most of the specific theories proposed to date in which a cataclysmic phenomenon mediated by the nervous system plays a role. An adequate analysis of this literature is not feasible in this review. The interested reader should peruse the references cited and those in the bibliographies of these papers to get some idea of the subject. Many possibilities exist, and in the opinion of this author, it is in this area of research that the ultimate answer to the triggering of the final death episode will be found. Sleep physiology of the developing infant is an area especially requiring intensive investigation.

'NEAR MISSES"

One of the most intriguing and potentially important areas of SIDS research involves infants found under circumstances that suggest they would otherwise have died of SIDS and who, after resuscitative efforts, either return to apparent normalcy or are left with neurological sequelae. We recently have observed two such cases in which the outcome was catastrophic. In one (case A-72-40), the 5-month-old son of a physician was found unresponsive in bed, previously having been in good health. Resuscitation was only partially successful, and severe anoxic encephalopathy supervened with death six days later. No anatomic cause for the apneic spell was found. Another case (A-72-49), a girl 7 years of age, had been found apneic at the age of 5 months, and despite cardiopulmonary response to resuscitative efforts she never regained consciousness in the subsequent years. At autopsy, extreme postanoxic encephalopathy was apparent. On the other hand, we have seen several dozen infants with similar spells who apparently were normal after similar spells, and in whom extensive observation in the hospital revealed no obvious anatomic or physiologic cause for the episode of sleep-associated apnea. An estimated 5% of our SIDS cases have a history of such a spell. Froggatt[20, 21] deserves credit for first pointing out that among cases of familial crib death one finds an increased incidence of apneic-cyanotic spells. In observed members of two families, multiple SIDS cases were associated with multiple spells, as described earlier.

Steinschneider[37] has done the most intensive investigation of this phenomenon and has published results of long-term monitoring of several such infants, including one tragic family in which five siblings died suddenly and unexpectedly, with findings typical of SIDS in several. Prolonged apneic spells most frequently were associated with the rapid eye movement (REM) phase of sleep and occurred more frequently when respiratory inflammation was present. It would appear probable that appropriate study of these important cases is one of the most promising leads to an understanding of SIDS. It was suggested by Landing, in the Second International Conference on Sudden Infant Death,[1] that the cause of SIDS eventually might be identified as also contributing to other disease states, such as brain damage of undetermined etiology in prematurely born and term infants. If such "spells" that lead to SIDS are capable of spontaneously aborting, then the morbidity of this phenomenon might be even greater than currently suspected.

The practical problem confronting the physician who is faced with a case of "near miss" is enormous. The parents understandably are panicked, and we usually have made a practice of hospitalizing such infants for several days of observation, partly in order to help the parents deal with their panic. The practical and philosophic problems involved in whether such infants should be sent home with a monitor are considerable, and no generalizations are justified on this at present. Whenever possible we have tried to avoid the use of monitors, as they represent a potentially significant interference with mother-child relationships. However, there are times when the emotional climate demands the use of a monitor. This is true especially in infants when such spells recur.

SYNTHESIS OF EPIDEMIOLOGIC AND PATHOLOGIC FINDINGS

Sudden infant death syndrome is characterized by at least two features which, to borrow Landing's apt term,[1] might be called *eligibility factors*—age and sleep. Virtually without exception, the victims of SIDS share these features. A number of other factors are frequent, but less constantly associated. These, which may influence the threshold or likelihood of occurrence of an SIDS event, may be termed *contingency factors*. Among the latter are *minor infections*, usually of the airway and usually viral, *prematurity* and *low socio-economic status*. The latter two, however, may not be independent variables, as described above. *Genetic factors*, possibly polygenic, may play some role in altering the threshold as well.

Death is *rapid* and *silent*. The pathological findings are minimal, and emphatically suggest that (1) an agonal episode of motor activity occurs and (2) this is attended by elevated intrathoracic negative pressure.

The above considerations have led us previously to discuss SIDS as probably representing a *final common pathway* in which similar agonal mechanisms of death are shared by the vast majority of cases. Theoretically, it is not necessary for the entire pathway to be identical in all cases. Probably many factors act in different ways to impinge on the sleeping infant and precipitate the lethal cataclysmic event. We have elsewhere discussed at some length[1] the possibility that the final pathway is *laryngospasm*. In this regard, the studies of Boushey *et al.*[61] are of interest, demonstrating the triggering of expiratory apneic reflexes with laryngeal irritation. Another pertinent observation was that of Baldridge, *et al.*,[62] in which laryngeal muscles were demonstrated to move in concurrence with eye musculature during REM sleep.

That death is mediated by a respiratory rather than a cardiac primary mechanism is suggested chiefly by the pathological evidence of increased negative pressure in the chest and the apparent silence of the death episode. This also is consistent with the observations of Steinschneider[37] and Stevens[63] on "near misses."

If one considers a susceptible, or "eligible" host, whose threshold for development of the lethal event is influenced by the balance of multiple factors, including constitutional determinants, specific developmental stage, internal and external chemical milieu, irritation from a minor airway inflammation, specific (?REM) stage of sleep, etc., then it is possible to conceptualize a more or less instantaneous catastrophic physiological event such as apnea with or without myotonic occlusion of the upper airway, leading to unexpected death in a manner consistent with the facts of SIDS as they are now understood. This latter catastrophic event is the "final common pathway" we have postulated to account for the tremendous similarity of most cases. The prevention of SIDS would seem to be prevention of the final common pathway, either directly which seems more certain, or indirectly by modifying the contingency factors affecting threshold, such as inflammatory disease.

It inherently seems probable that this relatively frequent phenomenon should not be limited to "healthy" infants, and almost certainly some SIDS mortality is undetectable because death by this mechanism happened to an infant with a potentially lethal disease. Several times annually we see deaths on the infant ward at Children's Hospital which occurred suddenly and were unobserved, and were attended by the same congestive pulmonary edema and intrathoracic petechiae as our SIDS cases. If such an infant has another underlying serious disease, it is of course that which appears on the death certificate, though one is tempted to ascribe the primary cause to SIDS.

Finally, it should be stated that series currently classified as SIDS, including our own, almost unquestionably include a number of individual disease entities, most of which remain to be discovered. It seems valid at this point in history to adopt the attitude of the "lumpers," but this attitude should not negate efforts to discriminate clinicopathologic subsets causing sudden unexpected infant death. For example, we have mentioned elsewhere Froggatt's[20] suggestion that at least some of the familial cases differ from the rest of the series by virtue of recurrent apneic spells in the history.

"To leave the family with a clear conscience is a duty secondary in importance only to saving the patient, and the physician should take all measures to be able to do this."[63]

There is at present no proven way to prevent crib death. However, it *is* possible to deal effectively with the family who has lost an infant to this tragedy.

Part of the syndrome of sudden infant death is the reaction of the person caring for the infant at the time—usually the mother. She automatically feels responsible for the death. Frequently added to this self-incrimination are direct or implied blame of others. Spouses, in-laws, neighbors and emergency personnel are often the source, but unfortunately physicians, police and the press add to the burden. Thus, the tragic loss of an infant is linked to dreadful reactions of guilt and blame which may persist for many years unless adequate and effective help is given early.[64-66] The principles of effective counseling are relatively simple and do not necessarily involve a great expenditure of time. The chief ingredients of effective management of this crucial problem are *knowledge* about SIDS, and *compassion* for the family. The latter may seem self-evident, but too often we have encountered situations where physicians, disapproving of one or another aspect of the parent's life style, have added enormously to the problem rather than preventing it. Far too often an infant death in a "nice" family is SIDS, but in other families an identical kind of death is interpreted as suffocation, neglect or deliberate infanticide. It unfortunately is not rare for parents of SIDS victims to be treated as suspected criminals in many communities.[67] We have dealt elsewhere in this work, especially in the "Epidemiology" section, with many of the factual aspects of effective counseling. Following are some additional points of importance.

1. *Autopsies are important and should be available for all infants, not only those whose families can afford them. However, they must be interpreted appropriately.* Many pathologists approach unexpected infant death with grave suspicion, and seem to harbor a fear of "missing" a case of infanticide. This fear of being wrong leads them to reluctance in using the term SIDS. This attitude is especially likely when the infant was illegitimate, or the family is poor, nonwhite, unkempt, intoxicated or has a criminal record. One noted forensic pathologist of our acquaintance certifies every such death that occurs in bed with an adult as "suffocation due to overlaying," yet he admits that cases occurring when the infant is alone in the crib are identical. It seems to us inherently illogical that SIDS cannot happen to a baby in bed with his parents!

2. The parents should be given an immediate diagnosis of SIDS.

Where the gross autopsy reveals no cause for death, with rare exceptions we make a diagnosis for counseling purposes of SIDS. The family cannot humanely be left in "limbo," waiting days or even weeks for results of microscopic, microbiologic, and toxicologic studies. I have made it a habit to call the family immediately upon completion of the autopsy, informing them of the diagnosis and telling them some of the facts of SIDS, emphasizing that it is not their fault, and that it is a specific disease entity and a very common cause of infant death. To learn it is *not their fault,* before erroneously based patterns of guilt and blame can become established, is an enormous relief to families. We have become convinced, through several years of using this approach, that the benefit vastly outweighs any disadvantages. If the subsequent laboratory findings reveal information such as unsuspected genetic disease of which the family should be made aware, they are, of course, so informed. Usually our approach in the latter instance is not to change their concept of the immediate cause of death: "Your baby did die of SIDS, but during our routine study of the case we found something about which you should know. . . ."

The person charged with making this immediate call will of course vary in other communities. In our hands it usually has proven best to have the author do this, as the person who actually did the postmortem examination carries considerable authority as to the nature and certainty of the diagnosis and is able to deal with questions relating to the postmortem findings with confidence. Probably in most communities another physician would assume this role, and in some jurisdictions a lay person, such as a coroner's secretary, has done an admirable job. The credentials of the person making this contact are less important than the rapidity and confidence with which a diagnosis is conveyed to the family.

3. *Printed material is of great help.* The Fact Sheet prepared and distributed by the National Foundation for Sudden Infant Death, Inc. (NFSID, address at the end of this section), is of great value in this regard. Copies are available from the NFSID upon request, and this or other similar material should be made available to families shortly after the death.

4. *Follow-up contacts are very useful.* We have for several years been involved in a pilot project, utilizing the County Visiting Nurse Service. After an annual indoctrination period in which the facts of SIDS are given to a group of skilled and specially motivated nurses, they undertake home visits, usually beginning about one week after the death, and continuing as long as the need exists. The grieving process is a slow one, and mothers derive great solace from learning that their adjustment problems are not unique—they are not "losing their minds." The nurses are often of value in advising how to deal with other siblings

in the home and may also be helpful at the time of birth of a subsequent child.

While the above formula has proven valuable in our hands, each community will draw upon its unique resources to develop its own mechanism for handling cases. The cornerstones of success are (1) a positive attitude toward the diagnosis of SIDS; (2) a compassionate attitude toward all families of SIDS victims; and (3) the availability of a mechanism for rapid routine notification and follow-up of families.

For those desiring additional information or printed materials, the National Foundation for Sudden Infant Death, Inc., 1501 Broadway, New York, N.Y. 10036 (phone 212-563-4630) stands ready to help. Among their various printed materials, the Fact Sheet and a recent brochure entitled, "The Subsequent Child," by Carolyn Szybist, R.N., herself the mother of an SIDS baby and a subsequent healthy child, are of especial value. The Foundation and its many chapters around the United States respond to requests from physicians and also directly from parents.

ACKNOWLEDGMENTS

The work presented herein is distilled from a team effort extending over seven years and involving many colleagues, of which a few can be mentioned here. Drs. Abraham B. Bergman and C. George Ray have collaborated in developing all data in this essay, and without their efforts over the years this paper could not have been written. Margaret Pomeroy, R.N., and Joyce Miller, R.N., also deserve special mention. Many volunteers in the community have been of particular help, among which Mrs. Mary Dore has been outstanding, as has Mrs. Sara Merkle.

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